TY B.Tech. – ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

PRACTICAL ASSIGNMENT TEMPLATE

1. Student Details

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2. Practical Details

Practical No.	6
Practical Title	Clustering Algorithms
Date Performed	24-10-2026
Faculty Name	Mrs. Anita Shingade

3. Problem Statement

To implement clustering algorithms on a real-world dataset (Wine chemical analysis data) in order to identify meaningful groups within the data without using class labels, by standardizing the numerical features, selecting the number of clusters using the Elbow method, Silhouette Score and Davies–Bouldin Index, and evaluating the quality of the clustering results using Silhouette Score, Davies–Bouldin Index, Calinski–Harabasz Index and visual inspection. The study also aims to compare K-Means, Hierarchical, Gaussian Mixture and DBSCAN clustering and to validate the discovered groups externally using the Adjusted Rand Index.

4. Objectives

1. To implement clustering algorithms on a real-world dataset to identify meaningful groups.
2. To preprocess numerical features using standardization so that all features contribute equally to distance calculations.
3. To determine a suitable number of clusters using the Elbow method, Silhouette Score and Davies–Bouldin Index.
4. To evaluate clustering quality using Silhouette Score, Davies–Bouldin Index, Calinski–Harabasz Index and visual inspection.
5. To compare K-Means, Hierarchical, Gaussian Mixture and DBSCAN clustering on the same data.
6. To interpret the discovered clusters and examine the stability of the selected model.

5. Expected Learning Outcomes

After completing this practical, the student will be able to:

1. Understand unsupervised learning and the working of partition-based, hierarchical, density-based and model-based clustering.
2. Implement clustering algorithms on real-world data using Python.
3. Evaluate clustering results using Silhouette Score, Davies–Bouldin Index, Calinski–Harabasz Index and visual inspection.
4. Compare clustering algorithms, interpret the clusters and justify the selected model.

6. Theory / Concept

Clustering

Clustering is an unsupervised learning technique that groups data points so that points in the same cluster are more similar to each other than to points in other clusters. No target label is used while fitting the model. For this application, the 13 chemical measurements of each wine are the input and the class label is kept aside only for external validation.

K-Means

K-Means partitions the data into k clusters by minimizing the within-cluster sum of squares (inertia):

$$J = \Sigma_k \Sigma_{x \in C_k} ||x - \mu_k||^2$$

where μ_k is the centroid of cluster C_k. It alternates between assigning each point to the nearest centroid and recomputing the centroids until the assignments stop changing.

Hierarchical (Agglomerative) Clustering

Every point starts as its own cluster and the two closest clusters are merged repeatedly. The linkage rule (Ward, complete, average or single) defines the distance between clusters. The result is visualized as a dendrogram that can be cut at any level to obtain k clusters.

DBSCAN

DBSCAN is a density-based method with two parameters: the neighbourhood radius ε (eps) and the minimum number of points MinPts. A point with at least MinPts neighbours within ε is a core point; clusters grow from core points, and points that belong to no dense region are labelled noise. It does not need k in advance.

Gaussian Mixture Model (GMM)

GMM models the data as a mixture of k Gaussian distributions:

$$p(x) = \Sigma_k \pi_k N(x \mid \mu_k, \Sigma_k)$$

and estimates π_k, μ_k and Σ_k with the Expectation–Maximization (EM) algorithm. Each point receives a probability of belonging to each cluster; the most probable cluster is used as its label.

Evaluation Metrics

Silhouette Score – for each point, $s(i) = (b(i) - a(i)) / \max(a(i), b(i))$, where a(i) is the mean distance to points of its own cluster and b(i) is the mean distance to the nearest other cluster. It lies in [−1, 1]; higher is better and negative values suggest a point may be in the wrong cluster.

Davies–Bouldin Index (DBI) – the average, over clusters, of the worst-case ratio of within-cluster scatter to between-cluster separation. Lower is better; 0 is the minimum.

Calinski–Harabasz Index (CH) – the ratio of between-cluster dispersion to within-cluster dispersion. Higher is better.

Adjusted Rand Index (ARI) – an external measure of agreement between the clusters and the known class labels, corrected for chance (about 0 for random labelling, 1 for perfect agreement). It is used here only for validation, never for fitting.

Application

Wine Cultivar Grouping using Chemical Analysis

Feature	Type	Use
alcohol	Numerical	Input
malic_acid	Numerical	Input
ash	Numerical	Input
alcalinity_of_ash	Numerical	Input
magnesium	Numerical	Input
total_phenols	Numerical	Input
flavanoids	Numerical	Input
nonflavanoid_phenols	Numerical	Input
proanthocyanins	Numerical	Input
color_intensity	Numerical	Input
hue	Numerical	Input
od280/od315_of_diluted_wines	Numerical	Input
proline	Numerical	Input
target (cultivar 0, 1, 2)	Categorical	Validation only

The problem is an unsupervised learning (clustering) problem because the aim is to discover groups from the 13 input features without using the class label.

K-Means is selected as the primary model because all features are continuous and standardized, it scales well, and its centroids give directly interpretable cluster profiles. Hierarchical clustering, Gaussian Mixture and DBSCAN are included to check whether the same structure is found by methods with different assumptions (nested, probabilistic and density-based).

7. Algorithm / Pseudocode

Clustering –

START
↓
Load Wine dataset
↓
Inspect data and missing values
↓
Separate features and class label (label kept only for validation)
↓
Standardize all features (StandardScaler)
↓
Select k using Elbow, Silhouette and Davies–Bouldin (k = 2 to 10)
↓
Train K-Means with the selected k
↓
Train Hierarchical clustering (Ward, complete, average, single)
↓
Train Gaussian Mixture Model
↓
Select eps using k-distance graph and grid search
↓
Train DBSCAN
↓
STOP

Evaluation and Validation –
START
↓
Calculate Silhouette, Davies–Bouldin and Calinski–Harabasz (noise excluded for DBSCAN)
↓
Calculate Adjusted Rand Index against class labels
↓
Reduce to 2 dimensions using PCA and plot clusters
↓
Plot silhouette plot and dendrogram
↓
Profile K-Means clusters using mean feature values
↓
Check K-Means stability over 20 random seeds
↓
Compare methods
↓
Analyze results
↓
STOP

08. Technology / Tools / Libraries Used

Technology / Tool	Purpose
Python	Programming
Google Colab / Jupyter Notebook	Development Environment
Pandas	Loading, inspecting and handling the Wine dataset
NumPy	Numerical computation
Scikit-learn	Dataset loading, scaling, clustering algorithms, PCA and evaluation metrics
SciPy	Hierarchical linkage and dendrogram
Matplotlib	Visualization

Dataset / Resources:

- 178 records
- 14 columns (13 numerical chemical features + 1 class label used only for validation)
- No missing values; all features numerical
- Source: UCI Wine dataset (loaded using sklearn.datasets.load_wine)
- Class sizes (not used for clustering): 59, 71 and 48 wines

Sample records (first five rows, selected columns):

alcohol	malic_acid	ash	flavanoids	color_intensity	proline	target
14.23	1.71	2.43	3.06	5.64	1065	0
13.20	1.78	2.14	2.76	4.38	1050	0
13.16	2.36	2.67	3.24	5.68	1185	0
14.37	1.95	2.50	3.49	7.80	1480	0
13.24	2.59	2.87	2.69	4.32	735	0

09. Implementation / GitHub Repository

Particular	Details
GitHub Repository URL	**https://github.com/Atharvfatale/Clustering-Algorithms.git**
Repository Name	**Clustering-Algorithms**
Branch Used	**main**
Commit ID / Latest Commit	**ef74c33fd60135e10886fab805167ee9a182eb5c**
Date of Last Commit	**24-10-2026**

10. Test Cases / Input and Output

Test Case	Input	Expected Output	Actual Output	Status
TC1	Load Wine dataset	Dataset loaded without missing values	Shape (178, 14); 0 missing values	Pass
TC2	Standardize the 13 features	Scaled feature matrix	Features scaled successfully	Pass
TC3	Select k for K-Means (k = 2 to 10) using Elbow, Silhouette and DBI	Optimal k identified	Best k = 3 (Silhouette 0.2849, DBI 1.3892)	Pass
TC4	Run K-Means, Hierarchical (Ward) and GMM with k = 3	Cluster label for every sample	3 clusters each, 0 noise points	Pass
TC5	Run DBSCAN (min_samples = 5, eps grid 1.5 to 3.5)	Clusters and noise identified	eps = 2.25: 2 clusters and 46 noise points	Pass
TC6	Evaluate with Silhouette, DBI, CH and ARI	Performance metrics generated	Metrics generated for all 7 configurations	Pass
TC7	Run K-Means with 20 different random seeds	Consistent clusters	Minimum ARI against reference run = 0.982	Pass

11. Results / Observations

The obtained results for Wine clustering are summarized below (k = 3 for K-Means, Hierarchical and GMM).

Method	Clusters	Noise	Silhouette	Davies–Bouldin	Calinski–Harabasz	ARI
K-Means	3	0	0.2849	1.3892	70.9400	0.8975
Hierarchical (Ward)	3	0	0.2774	1.4186	67.6475	0.7899
Hierarchical (Complete)	3	0	0.2038	1.8961	48.9898	0.5771
Hierarchical (Average)	3	0	0.1575	1.0299	4.0314	−0.0054
Hierarchical (Single)	3	0	0.1827	0.9105	4.0617	−0.0068
Gaussian Mixture	3	0	0.2844	1.3938	70.6878	0.8804
DBSCAN (eps 2.25)	2	46	0.3371*	1.1712*	71.0774*	0.3962

** DBSCAN metrics are computed on the 132 non-noise points only, so they are not directly comparable with the other rows.*

Parameter	Value
Number of samples / features	178 / 13
Scaling	StandardScaler
Selected k (Silhouette maximum)	3
K-Means settings	n_init = 10, random_state = 42
DBSCAN settings	eps = 2.25, min_samples = 5
K-Means cluster sizes	65, 51, 62
K-Means samples with negative silhouette	7 of 178
PCA variance explained (PC1, PC2)	36.20%, 19.21% (total 55.41%)
K-Means stability (20 seeds, minimum ARI)	0.982

K-Means cluster profile (mean values in original units):

Feature	Cluster 0 (65)	Cluster 1 (51)	Cluster 2 (62)
alcohol	12.25	13.13	13.68
malic_acid	1.90	3.31	2.00
ash	2.23	2.42	2.47
alcalinity_of_ash	20.06	21.24	17.46
magnesium	92.74	98.67	107.97
total_phenols	2.25	1.68	2.85
flavanoids	2.05	0.82	3.00
nonflavanoid_phenols	0.36	0.45	0.29
proanthocyanins	1.62	1.15	1.92
color_intensity	2.97	7.23	5.45
hue	1.06	0.69	1.07
od280/od315_of_diluted_wines	2.80	1.70	3.16
proline	510.17	619.06	1100.23

Observations

- K-Means, Ward Hierarchical and Gaussian Mixture all found 3 clusters of similar quality (Silhouette 0.2849, 0.2774 and 0.2844).
- k = 3 was supported by all three internal criteria: highest Silhouette (0.2849), lowest Davies–Bouldin (1.3892) and highest Calinski–Harabasz (70.94) among k = 2 to 10. The Elbow curve bends gradually, so it gives only weak support on its own.
- A mean Silhouette of about 0.28 indicates moderate-to-weak separation: the clusters are real but overlap. 7 of 178 samples (3.9%) have a negative silhouette in K-Means.
- K-Means clusters agree strongly with the known wine classes (ARI = 0.8975) although the labels were never used for fitting.
- DBSCAN reports the best Silhouette (0.3371) and DBI (1.1712), but it found only 2 clusters (90 and 42 points) and labelled 46 of 178 points (25.8%) as noise; its ARI is only 0.3962.
- Average and single linkage give the lowest DBI (1.0299 and 0.9105) but place 174 of 178 points in one cluster (sizes 174/1/3 and 174/3/1), with ARI close to 0 and CH close to 4.
- K-Means is stable: across 20 different random seeds the minimum ARI against the reference run was 0.982.
- The 2D PCA view explains 55.41% of the variance and shows three groups with some overlap at the boundaries.

Screenshots / Graphs / Output:

```
Dataset Shape: (178, 14)
Missing values: 0
Features: ['alcohol', 'malic_acid', 'ash', 'alcalinity_of_ash', 'magnesium', 'total_phenols', 'flavanoids',
'nonflavanoid_phenols', 'proanthocyanins', 'color_intensity', 'hue', 'od280/od315_of_diluted_wines', 'proline']
Class sizes (not used for clustering): [59, 71, 48]
```

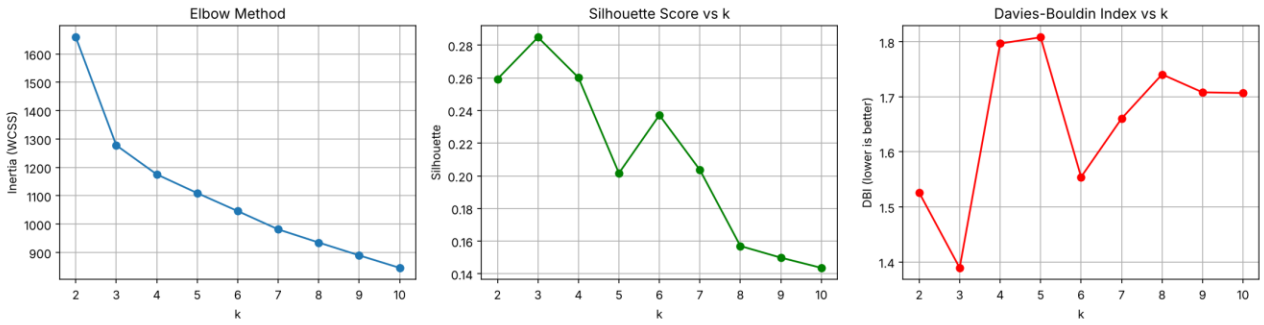


Figure 1: Elbow method, Silhouette score and Davies–Bouldin index for k = 2 to 10

```
K SELECTION
 k    Inertia  Silhouette  Davies_Bouldin  Calinski_Harabasz
 2 1658.7589      0.2593          1.5260            69.5233
 3 1277.9285      0.2849          1.3892            70.9400
 4 1175.4283      0.2602          1.7969            56.1814
 5 1109.5127      0.2016          1.8083            46.9522
 6 1046.0023      0.2372          1.5544            41.7008
 7  981.5952      0.2036          1.6606            38.6855
 8  935.2012      0.1570          1.7409            35.8053
 9  889.8929      0.1499          1.7083            33.8066
10  845.8952      0.1436          1.7070            32.3972

Best k by silhouette: 3
```

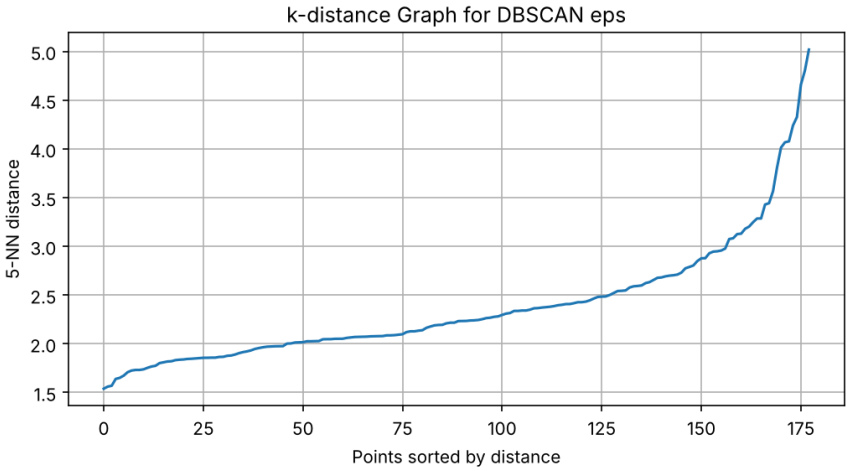


Figure 2: k-distance graph (5th nearest neighbour) used to choose the DBSCAN eps range

```
DBSCAN GRID (min_samples=5)
 eps  Clusters  Noise  Silhouette
1.50         0    178         NaN
1.75         6    132      0.3129
2.00         5     85      0.2405
2.25         2     46      0.3371
2.50         1     24         NaN
2.75         1     17         NaN
3.00         1     11         NaN
3.25         1      8         NaN
3.50         1      6         NaN
Selected eps: 2.25
```

```
============================================================
FINAL COMPARISON
============================================================
                Method  Clusters  Noise  Silhouette  Davies_Bouldin  Calinski_Harabasz     ARI
               K-Means         3      0      0.2849          1.3892            70.9400  0.8975
    Hierarchical (ward)        3      0      0.2774          1.4186            67.6475  0.7899
Hierarchical (complete)        3      0      0.2038          1.8961            48.9898  0.5771
 Hierarchical (average)        3      0      0.1575          1.0299             4.0314 -0.0054
  Hierarchical (single)        3      0      0.1827          0.9105             4.0617 -0.0068
      Gaussian Mixture         3      0      0.2844          1.3938            70.6878  0.8804
                DBSCAN         2     46      0.3371          1.1712            71.0774  0.3962
```

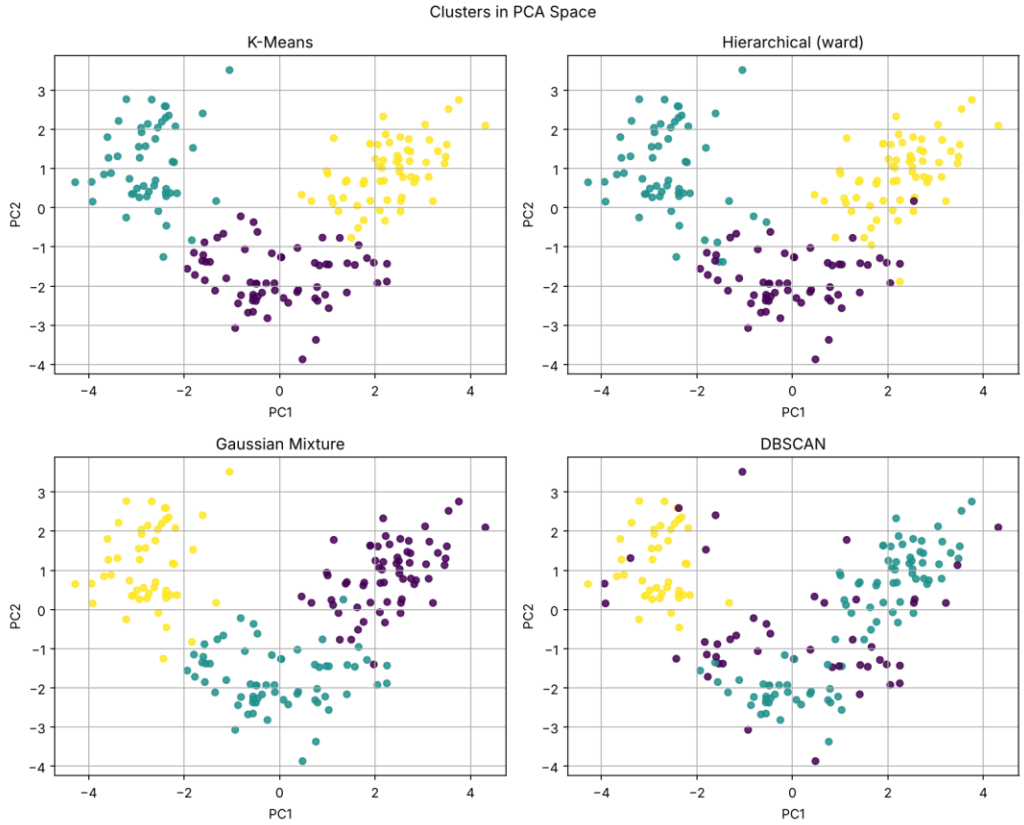


Figure 3: Cluster assignments of the four methods shown in PCA space (colours are arbitrary labels)

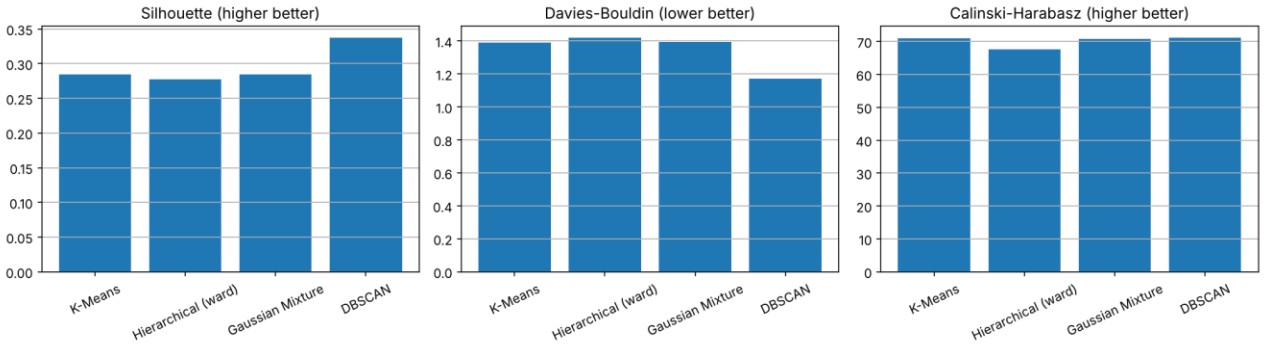


Figure 4: Silhouette, Davies–Bouldin and Calinski–Harabasz comparison of the four main methods

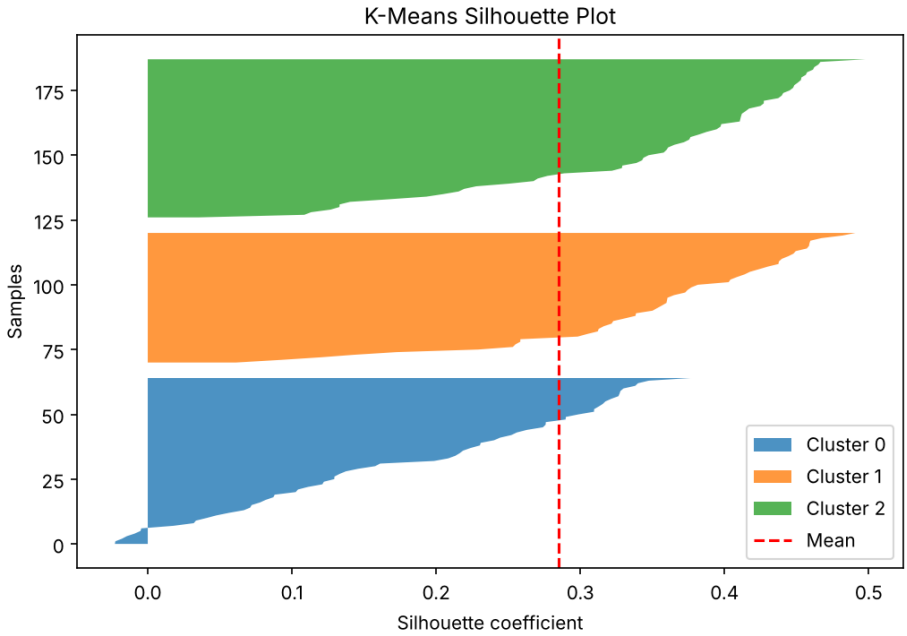


Figure 5: K-Means silhouette plot (dashed line = mean silhouette 0.2849)

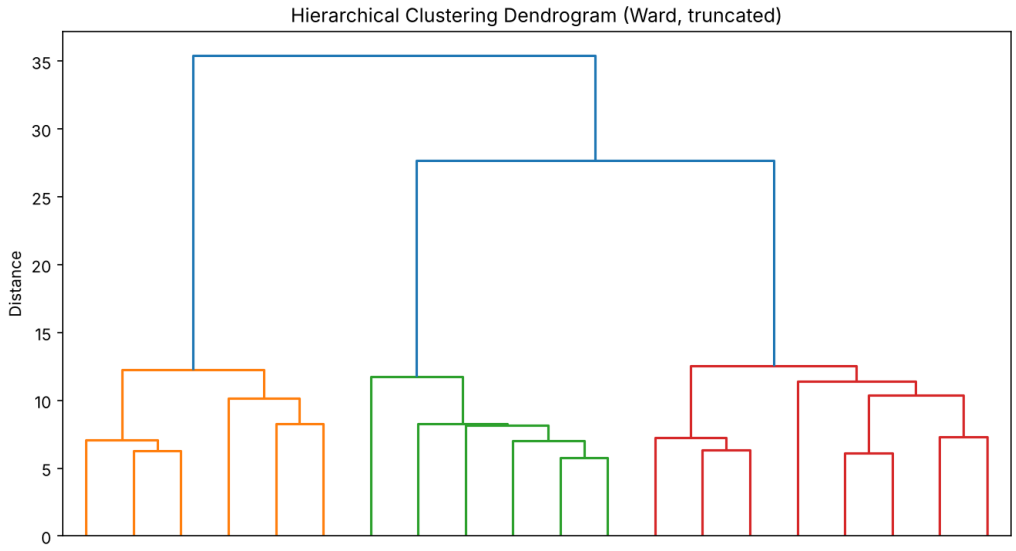


Figure 6: Ward dendrogram (truncated to the last 20 merges)

```
PCA explained variance (PC1, PC2): [0.362, 0.1921] sum= 0.5541
K-Means samples with negative silhouette: 7


K-Means cluster profile (original units):
Cluster                          0       1        2
alcohol                      12.25   13.13    13.68
malic_acid                    1.90    3.31     2.00
ash                           2.23    2.42     2.47
alcalinity_of_ash            20.06   21.24    17.46
magnesium                    92.74   98.67   107.97
total_phenols                 2.25    1.68     2.85
flavanoids                    2.05    0.82     3.00
nonflavanoid_phenols          0.36    0.45     0.29
proanthocyanins               1.62    1.15     1.92
color_intensity               2.97    7.23     5.45
hue                           1.06    0.69     1.07
od280/od315_of_diluted_wines  2.80    1.70     3.16
proline                     510.17  619.06  1100.23
Cluster sizes: [65, 51, 62]


K-Means seed stability (min ARI vs seed 0 over 20 seeds): 0.982
```

12. Performance Evaluation / Analysis

Parameter	K-Means	Hierarchical (Ward)	Gaussian Mixture	DBSCAN
Silhouette Score	0.2849	0.2774	0.2844	0.3371*
Davies–Bouldin Index	1.3892	1.4186	1.3938	1.1712*
Calinski–Harabasz Index	70.94	67.65	70.69	71.08*
ARI (vs class labels)	0.8975	0.7899	0.8804	0.3962
Clusters / Noise points	3 / 0	3 / 0	3 / 0	2 / 46
Needs k in advance	Yes	Yes (to cut tree)	Yes	No
Handles noise / outliers	No	No	No	Yes
Cluster shape assumed	Spherical	Compact (Ward)	Elliptical	Arbitrary
Type of assignment	Hard	Hard	Soft (probabilities)	Hard + noise

** computed after excluding 46 noise points; not directly comparable.*

Discussion

- **Model selection:** K-Means with k = 3 is selected because three independent internal criteria agree on k = 3, it scores best among the three full-coverage methods, and it gives interpretable centroids. Ward and GMM reach nearly the same quality, so the structure is not an artefact of K-Means.
- **Internal vs external agreement:** the Silhouette Score is modest (0.2849) but ARI is high (0.8975). The wine groups overlap in 13-dimensional space, which lowers silhouette, yet the partition still matches the true cultivars well. ARI is only available because labels exist; in a real unlabelled problem only the internal metrics could be used.
- **Cluster interpretation:** Cluster 0 (65 wines) has the lowest alcohol (12.25), colour intensity (2.97) and proline (510.17). Cluster 1 (51 wines) has the highest malic acid (3.31) and colour intensity (7.23) and the lowest flavanoids (0.82) and hue (0.69). Cluster 2 (62 wines) has the highest proline (1100.23), flavanoids (3.00) and total phenols (2.85).
- **DBSCAN:** its better-looking metrics are misleading because 25.8% of the points are removed as noise and the metrics are calculated on the rest. With 178 points in 13 dimensions, small eps values mark most points as noise (eps 1.5 gives 178 noise points) and larger values merge everything into one cluster (eps 2.5 and above), so no eps gives three balanced clusters.

- **Metric disagreement:** average and single linkage have the best DBI but are degenerate (one cluster holds 174 of 178 points). DBI alone would have selected a useless result, which is why several metrics, ARI and visual inspection were used together.
- **Visual inspection:** the PCA plot shows three groups with boundary overlap. PCA retains only 55.41% of the variance, so the 2D picture is approximate.
- **Stability:** K-Means is repeatable (minimum ARI 0.982 over 20 seeds).

Limitations, Ethical Considerations and Future Learning

- The dataset is small (178 samples) and is a well-known benchmark, so the results may not generalize to larger or noisier data.
- DBSCAN parameters were chosen by maximum Silhouette over a grid, which favoured a 2-cluster solution with many noise points; a constraint on the noise fraction would be a better criterion.
- The Wine data contains no personal information. When clustering is applied to people (customers, patients), clusters can encode bias and lead to profiling, so they should be treated as descriptive groups, not facts, and checked for fairness.
- The class label was excluded from fitting and used only for validation, and fixed random seeds make the experiment reproducible.
- Future learning: HDBSCAN and OPTICS, spectral clustering, feature selection, UMAP and bootstrap-based stability analysis.

Suggested Improvements

- Apply feature selection or PCA before clustering to reduce uninformative dimensions.
- Use the gap statistic or bootstrap stability in addition to Silhouette and DBI for choosing k.
- Compare HDBSCAN, OPTICS and Spectral Clustering.
- Validate the approach on a larger real-world dataset such as customer segmentation data.

13. Conclusion

The Wine chemical analysis dataset was successfully used to identify groups in a real-world dataset without using class labels. After standardizing the 13 features, K-Means, Hierarchical, Gaussian Mixture and DBSCAN clustering were implemented and evaluated using Silhouette Score, Davies–Bouldin Index, Calinski–Harabasz Index and visual inspection.

K-Means with k = 3 was selected as the final model. It achieved a Silhouette Score of 0.2849, a Davies–Bouldin Index of 1.3892 and a Calinski–Harabasz Index of 70.94, with three clusters of 65, 51 and 62 wines. The Silhouette value shows that the clusters overlap, but the external ARI of 0.8975 and the minimum seed-to-seed ARI of 0.982 show that the grouping is meaningful and stable.

Ward Hierarchical clustering and Gaussian Mixture gave similar results, whereas DBSCAN labelled 25.8% of the points as noise and average/single linkage produced one dominant cluster, showing that a single internal metric can be misleading. Therefore the experiment confirms that suitable clustering depends on the algorithm assumptions and that results must be judged using several metrics together with visual inspection.

14. Assignment Questions

Q1. Why must features be scaled before applying clustering algorithms?

Answer : K-Means, Ward hierarchical clustering and DBSCAN depend on Euclidean distance, so a feature with a large numeric range dominates the distance. In the Wine data, proline in the first five rows lies

between 735 and 1480, whereas hue lies between 0.86 and 1.05. Without scaling, proline would almost completely decide the clusters. StandardScaler gives every feature zero mean and unit variance so all features contribute equally.

Q2. How do the Silhouette Score and Davies–Bouldin Index evaluate clustering, and how was k selected?

Answer : The Silhouette Score compares, for every point, the distance to its own cluster with the distance to the nearest other cluster; values near 1 indicate compact, well-separated clusters and negative values indicate misassigned points. The Davies–Bouldin Index measures the average similarity between each cluster and its most similar cluster, so lower values are better. K-Means was run for k = 2 to 10; k = 3 gave the maximum Silhouette (0.2849) and the minimum DBI (1.3892), so k = 3 was selected.

Q3. When is DBSCAN preferred over K-Means, and how did it perform here?

Answer : DBSCAN is preferred when clusters have arbitrary shapes, the number of clusters is unknown, or the data contains outliers that should be labelled as noise. In this experiment the Wine clusters are overlapping and roughly compact in 13 dimensions, so DBSCAN performed poorly: it found 2 clusters, marked 46 points as noise and had an ARI of only 0.3962, compared with 0.8975 for K-Means.

15. Implementation / Source Code

```
import json
import numpy as np
import pandas as pd
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt

from sklearn.datasets import load_wine
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans, AgglomerativeClustering, DBSCAN
from sklearn.mixture import GaussianMixture
from sklearn.neighbors import NearestNeighbors
from sklearn.metrics import (silhouette_score, davies_bouldin_score,
                             calinski_harabasz_score, adjusted_rand_score)
from scipy.cluster.hierarchy import dendrogram, linkage

RS = 42
out = {}

# 1. LOAD DATASET
data = load_wine(as_frame=True)
df = data.frame.copy()
print("Dataset Shape:", df.shape)
print("\nFirst 5 Rows:")
print(df.head().to_string())
print("\nMissing values:", int(df.isnull().sum().sum()))

# 2. SEPARATE FEATURES (labels kept ONLY for external validation)
y_true = df["target"].values
```

```
X = df.drop(columns=["target"])
print("\nFeatures:", X.columns.tolist())
print("Class sizes (not used for clustering):", np.bincount(y_true).tolist())

# 3. SCALING
X_scaled = StandardScaler().fit_transform(X)

# 4. CHOOSING K: elbow + silhouette + Davies-Bouldin for k = 2..10
ks = list(range(2, 11))
inertia, sil, dbi, chi = [], [], [], []
for k in ks:
    km = KMeans(n_clusters=k, n_init=10, random_state=RS).fit(X_scaled)
    inertia.append(km.inertia_)
    sil.append(silhouette_score(X_scaled, km.labels_))
    dbi.append(davies_bouldin_score(X_scaled, km.labels_))
    chi.append(calinski_harabasz_score(X_scaled, km.labels_))
sel = pd.DataFrame({"k": ks, "Inertia": inertia, "Silhouette": sil,
                    "Davies_Bouldin": dbi, "Calinski_Harabasz": chi})
print("\nK SELECTION")
print(sel.round(4).to_string(index=False))
best_k = int(ks[int(np.argmax(sil))])
print("\nBest k by silhouette:", best_k)
out["best_k"] = best_k
out["k_selection"] = sel.round(4).to_dict(orient="records")

fig, ax = plt.subplots(1, 3, figsize=(15, 4))
ax[0].plot(ks, inertia, "o-"); ax[0].set_title("Elbow Method"); ax[0].set_xlabel("k");
ax[0].set_ylabel("Inertia (WCSS)")
ax[1].plot(ks, sil, "o-", color="green"); ax[1].set_title("Silhouette Score vs k"); ax[1].set_xlabel("k");
ax[1].set_ylabel("Silhouette")
ax[2].plot(ks, dbi, "o-", color="red"); ax[2].set_title("Davies-Bouldin Index vs k");
ax[2].set_xlabel("k"); ax[2].set_ylabel("DBI (lower is better)")
for a in ax: a.grid(True)
plt.tight_layout(); plt.savefig("fig1_k_selection.png", dpi=150); plt.close()

# 5. MODELS
results = []
labels_store = {}

def evaluate(name, labels, extra=""):
    mask = labels != -1
    n_clusters = len(set(labels[mask]))
    n_noise = int((~mask).sum())
    if n_clusters >= 2:
        s = silhouette_score(X_scaled[mask], labels[mask])
        d = davies_bouldin_score(X_scaled[mask], labels[mask])
        c = calinski_harabasz_score(X_scaled[mask], labels[mask])
    else:
        s = d = c = float("nan")
```

```
    ari = adjusted_rand_score(y_true, labels)
    results.append({"Method": name, "Clusters": n_clusters, "Noise": n_noise,
                    "Silhouette": s, "Davies_Bouldin": d,
                    "Calinski_Harabasz": c, "ARI": ari, "Note": extra})
    labels_store[name] = labels

# 5a K-Means
km = KMeans(n_clusters=best_k, n_init=10, random_state=RS).fit(X_scaled)
evaluate("K-Means", km.labels_)

# 5b Hierarchical (Ward) + linkage comparison
for link in ["ward", "complete", "average", "single"]:
    ag = AgglomerativeClustering(n_clusters=best_k, linkage=link).fit(X_scaled)
    evaluate(f"Hierarchical ({link})", ag.labels_)

# 5c GMM
gmm = GaussianMixture(n_components=best_k, covariance_type="full", n_init=5,
random_state=RS).fit(X_scaled)
evaluate("Gaussian Mixture", gmm.predict(X_scaled))

# 5d DBSCAN: eps from k-distance graph (min_samples = 2*dim rule is too strict for 178 pts -> use 5)
min_samples = 5
nn = NearestNeighbors(n_neighbors=min_samples).fit(X_scaled)
kdist = np.sort(nn.kneighbors(X_scaled)[0][:, -1])
plt.figure(figsize=(7, 4)); plt.plot(kdist); plt.xlabel("Points sorted by distance");
plt.ylabel(f"{min_samples}-NN distance")
plt.title("k-distance Graph for DBSCAN eps"); plt.grid(True); plt.tight_layout()
plt.savefig("fig2_kdistance.png", dpi=150); plt.close()

grid = []
for eps in np.arange(1.5, 3.55, 0.25):
    lab = DBSCAN(eps=float(eps), min_samples=min_samples).fit_predict(X_scaled)
    m = lab != -1
    nc = len(set(lab[m]))
    s = silhouette_score(X_scaled[m], lab[m]) if nc >= 2 else float("nan")
    grid.append({"eps": round(float(eps), 2), "Clusters": nc, "Noise": int((~m).sum()), "Silhouette": s})
grid_df = pd.DataFrame(grid)
print("\nDBSCAN GRID (min_samples=5)")
print(grid_df.round(4).to_string(index=False))
out["dbscan_grid"] = grid_df.round(4).fillna("NA").to_dict(orient="records")
valid = grid_df[(grid_df.Clusters >= 2) & (grid_df.Noise <= 0.3 * len(X_scaled))]
if len(valid):
    best_eps = float(valid.sort_values("Silhouette", ascending=False).iloc[0]["eps"])
else:
    best_eps = float(grid_df.iloc[len(grid_df) // 2]["eps"])
print("Selected eps:", best_eps)
out["dbscan_eps"] = best_eps
db_lab = DBSCAN(eps=best_eps, min_samples=min_samples).fit_predict(X_scaled)
evaluate("DBSCAN", db_lab, f"eps={best_eps}, min_samples={min_samples}")
```

```
res = pd.DataFrame(results)
print("\n" + "=" * 60)
print("FINAL COMPARISON")
print("=" * 60)
print(res.drop(columns=["Note"]).round(4).to_string(index=False))
res.round(4).to_csv("clustering_results.csv", index=False)
out["results"] = res.round(4).fillna("NA").to_dict(orient="records")

# 6. PCA VISUALISATION
pca = PCA(n_components=2, random_state=RS)
Z = pca.fit_transform(X_scaled)
ev = pca.explained_variance_ratio_
print("\nPCA explained variance (PC1, PC2):", ev.round(4).tolist(), "sum=", round(float(ev.sum()), 4))
out["pca_var"] = [round(float(v), 4) for v in ev]

show = ["K-Means", "Hierarchical (ward)", "Gaussian Mixture", "DBSCAN"]
fig, ax = plt.subplots(2, 2, figsize=(11, 9))
for a, name in zip(ax.ravel(), show):
    lab = labels_store[name]
    sc = a.scatter(Z[:, 0], Z[:, 1], c=lab, cmap="viridis", s=25, alpha=0.85)
    a.set_title(name); a.set_xlabel("PC1"); a.set_ylabel("PC2"); a.grid(True)
plt.suptitle("Clusters in PCA Space"); plt.tight_layout()
plt.savefig("fig3_pca_clusters.png", dpi=150); plt.close()

# 7. SILHOUETTE PLOT FOR K-MEANS
from sklearn.metrics import silhouette_samples
sv = silhouette_samples(X_scaled, km.labels_)
plt.figure(figsize=(7, 5))
y0 = 0
for c in range(best_k):
    v = np.sort(sv[km.labels_ == c])
    plt.fill_betweenx(np.arange(y0, y0 + len(v)), 0, v, alpha=0.8, label=f"Cluster {c}")
    y0 += len(v) + 5
plt.axvline(silhouette_score(X_scaled, km.labels_), color="red", linestyle="--", label="Mean")
plt.xlabel("Silhouette coefficient"); plt.ylabel("Samples"); plt.title("K-Means Silhouette Plot")
plt.legend(); plt.tight_layout(); plt.savefig("fig4_silhouette_plot.png", dpi=150); plt.close()
out["kmeans_neg_sil"] = int((sv < 0).sum())
print("K-Means samples with negative silhouette:", out["kmeans_neg_sil"])

# 8. DENDROGRAM
Zl = linkage(X_scaled, method="ward")
plt.figure(figsize=(9, 5))
dendrogram(Zl, truncate_mode="lastp", p=20, no_labels=True)
plt.title("Hierarchical Clustering Dendrogram (Ward, truncated)"); plt.ylabel("Distance")
plt.tight_layout(); plt.savefig("fig5_dendrogram.png", dpi=150); plt.close()

# 9. METRIC BAR CHART
main = res[res.Method.isin(show)]
```

```
fig, ax = plt.subplots(1, 3, figsize=(14, 4))
for a, col, ttl in zip(ax, ["Silhouette", "Davies_Bouldin", "Calinski_Harabasz"],
                       ["Silhouette (higher better)", "Davies-Bouldin (lower better)", "Calinski-Harabasz (higher better)"]):
    a.bar(main.Method, main[col]); a.set_title(ttl); a.tick_params(axis="x", rotation=25); a.grid(axis="y")
plt.tight_layout(); plt.savefig("fig6_metric_comparison.png", dpi=150); plt.close()

# 10. CLUSTER PROFILE (K-Means), original units
prof = X.copy(); prof["Cluster"] = km.labels_
profile = prof.groupby("Cluster").mean().round(2)
print("\nK-Means cluster profile (original units):")
print(profile.T.to_string())
out["cluster_sizes"] = np.bincount(km.labels_).tolist()
print("Cluster sizes:", out["cluster_sizes"])
profile.to_csv("kmeans_profile.csv")

# 11. STABILITY: K-Means with 20 different seeds
aris = []
base = KMeans(n_clusters=best_k, n_init=10, random_state=0).fit(X_scaled).labels_
for sd in range(1, 21):
    l = KMeans(n_clusters=best_k, n_init=10, random_state=sd).fit(X_scaled).labels_
    aris.append(adjusted_rand_score(base, l))
out["stability_min_ari"] = round(float(min(aris)), 4)
print("\nK-Means seed stability (min ARI vs seed 0 over 20 seeds):", out["stability_min_ari"])

with open("results.json", "w") as f:
    json.dump(out, f, indent=2, default=str)
print("\nDone.")
```

16. Student Declaration

I hereby declare that the above practical work has been completed by me and that the implementation, analysis, and answers are my own work.

Student Name	PRN	Signature
Atharv Jeevan Fatale	202401070076	Atharvfatale